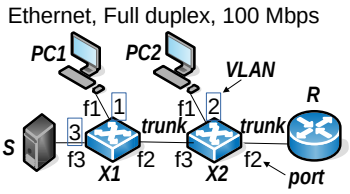
Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	19/06/2024	Primavera 2024
Name	Surname	Group	DNI

Duration: 3h. The quiz will be collected in 30 minutes. Answer the problems in the same sheet.

Quiz (2points) Mark the correct answers. All the questions are multi-answer: all combinations are possible (from all false to all true); and they score half if there is an error, 0 if more.

1. In the network in the figure, estimate the value of the optimal window measured in segments for a TCP connection between PC2 and S. Assume that acks have size equal to 0 and that the delay of propagation on the links is equal to 0. Note that the transmission time is not zero.



   ☐ 5 ☐ It depends on R's buffer size ☐ 10 ☐ 1 ☐ 2 ☐ 6 ☐ 3 ☐ 12

2. Given the network in the figure and the assumptions above, estimate what the throughput in Mbps will be for a connection using a window equal to 1 segment (ie, stop-and-wait) between PC2 and S.

   ☐ 20 ☐ 10 ☐ 33,33 ☐ 100 ☐ 16,66 ☐ 50 ☐ 8,33

3. In the network in the figure PC1 and PC2 send data at the maximum rate allowed by the network with TCP to S. Say on which ports the switches are likely to send pause frames. Notation: switch, port.

   ☐ X1, f1 ☐ X2, f1 ☐ X1, f3 ☐ X1, f2 ☐ X2, f2 ☐ X2, f3

4. In the network in the figure all the ARP tables are empty. PC1 pings the IP address of S. Tell whether the following IP addresses will be in the ARP tables when PC1 receives a response. There is only the traffic generated by the ping. Notation: device, IP_{device}; where IP_{device} is the IP address of some interface on the device.

   ☐ PC2, IP_R ☐ S, IP_R ☐ PC2, IP_S ☐ PC1, IP_S ☐ S, IP_{PC1} ☐ PC1, IP_R ☐ R, IP_S ☐ R, IP_{PC1}

5. In the network in the figure all the MAC tables (switch forwarding table) and ARP tables are empty. PC1 pings the IP address of S. Tell whether the following MAC addresses will be in the MAC tables when PC1 receives a response. There is only the traffic generated by the ping. Notation: switch, VLAN, port, device-MAC address (M_{device}).

   ☐ X2, 3, f2, M_S ☐ X2, 2, f1, M_{PC2} ☐ X1, 3, f2, M_R ☐ X2, 3, f2, M_R ☐ X2, 1, f3, M_{PC1} ☐ X1, 2, f2, M_R ☐ X1, 1, f2, M_R

6. In the network in the figure, say at least how many IPs must be assigned to device interfaces so that there is connectivity between PC1, PC2, and S:

   ☐ 3 ☐ 4 ☐ 6 ☐ 8 ☐ 7 ☐ 5

7. State which characteristics are true of a LAN:
   - ☐ In a Wifi AP there can be PCs connected in Half Duplex and Full Duplex mode
   - ☐ Ethernet frames carry two addresses
   - ☐ One of the addresses in WiFi frames is the BSSID
   - ☐ On a switch there can be PCs connected in Half Duplex and Full Duplex mode
   - ☐ WiFi frames carry three or more addresses

8. Say if the headers of the following protocols contain the indicated fields:
   - ☐ IP, don't fragment flag
   - ☐ TCP, PUSH flag
   - ☐ IP, protocol field
   - ☐ UDP, PUSH flag
   - ☐ UDP, TTL field

9. Say which devices use the given protocols:
   - ☐ Routers, Spanning Tree Protocol (STP)
   - ☐ Routers, Internet Control Message Protocol (ICMP)
   - ☐ Switches, Spanning Tree Protocol (STP)
   - ☐ AP WiFi, Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
   - ☐ Hubs, Carrier Sense Multiple Access with Collision Detection (CSMA/CD)
   - ☐ Switches, Internet Control Message Protocol (ICMP)

10. Say which statements are true about TCP/UDP:
    - ☐ Both protocols have a checksum field to detect errors
    - ☐ In both protocols, ACKs are sent to confirm the correct receipt of information
    - ☐ The "advertised window" field in the TCP header varies depending on network congestion
    - ☐ The ports that identify the client and server socket are always sent in the header of the two protocols

11. Say which of the following statements about the email application are true:
    - ☐ To send the message to the mail server a client can use HTTP or SMTP
    - ☐ The mail client must request an MX time record resource from the name server before sending the message
    - ☐ To be able to send messages with binary content it needs to use MIME
    - ☐ To read mail, the client can download messages from the server using the SMTP protocol

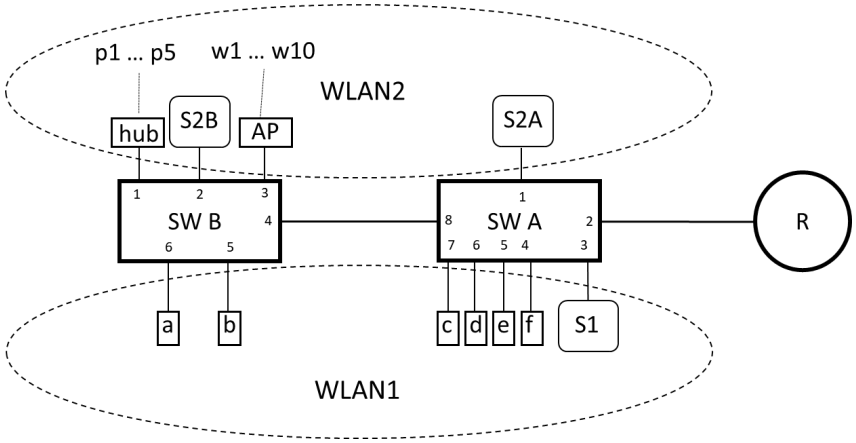
12. Say which statements are true about HTTP:
    - ☐ In the first line of the Response message there is a code that indicates if there is an error
    - ☐ The data sent by the server must be encoded in ASCII
    - ☐ One of the fields in the Query and Response message header can be Content-Type
    - ☐ In the first line of the Query message there is the method (GET, POST...)
    - ☐ One of the header fields of the Response message can be Reset to abort the connection

Examen final. Xarxes de Computadors (XC), Grau en Enginyeria Informàtica		19/06/2024	Spring 2024
NAME (UPPER CASE):	LAST NAME (UPPER CASE):	GROUP:	DNI/NIE:

Answer on the same sheet.

Problem 1 (1.75 points)

The figure shows a network with two VLANs and two Ethernet switches (SWA and SWB). VLAN1 includes devices a, b, c, d, e, f and server S1. VLAN 2 includes servers S2A and S2B, a "*hub*" with 5 devices (pi) and a WiFi access point at 120Mbps with 10 devices (wi). The efficiency of the "*hub*" and the WiFi access point is 80%. Control flow is activated in both switches and all links are Fast Ethernet (100Mbps). All connections between client and server are TCP with sustained traffic at the maximum achievable rate.



For each of the following scenarios identify which ports are a bottleneck and activate flow control (notation: SWx-port) and the effective transmission rate the devices may achieve.

a) (0.25) All devices in VLAN1 send information to S1. No other communications are present.
Port/ports with bottleneck:
Effective throughput in Mbps for a: b: c: d: e: f:

b) (0.25) S1 sends information to all the devices in VLAN1. No other communications are present.
Port/ports with bottleneck:
Effective throughput in Mbps for a: b: c: d: e: f:

c) (0.35) All devices in VLAN1 send information to S1 and at the same time S1 sends information to all the devices in VLAN1. No other communications are present.
Port/ports with bottleneck:
Effective transmission throughput in Mbps for a: b: c: d: e: f:
Effective reception throughput in Mbps for a: b: c: d: e: f:

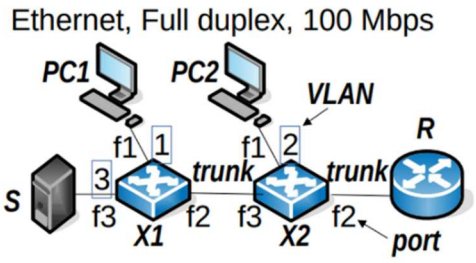
d) (0.25) S1 sends information to all the devices in VLAN2. No other communications are present.
Port/ports with bottleneck:
Effective throughput in Mbps for p: w: S2A: S2B:

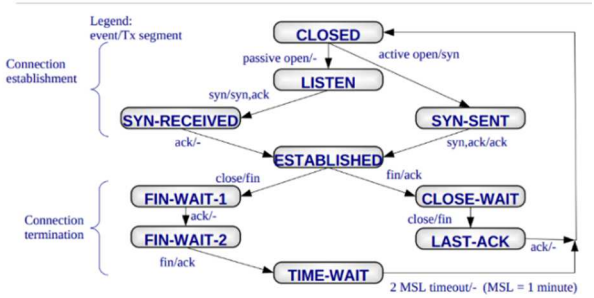
e) (0.25) All devices in VLAN2 send information to S1. No other communications are present.
Port/ports with bottleneck:
Effective throughput in Mbps for p: w: S2A: S2B:

f) (0.40) All devices in VLAN2 send information to S1 and at the same time S1 sends information to all the devices in VLAN2. No other communications are present.
Port/ports with bottleneck:
Effective transmission throughput in Mbps for p: w: S2A: S2B:
Effective reception throughput in Mbps for p: w: S2A: S2B:

Xarxes de Computadors (XC) exam, GEI		19/06/2024	Spring 2024
NAME (CAPITAL LETTERS):	SURNAME (CAP LETTERS):	GROUP:	ID:

Prob. 2 (2pt) In the network in the figure (it is the same as the test network) we have two switches (X1 and X2), two PCs in different VLANs (PC1 in VLAN1 and PC2 in VLAN2), a server (S) in VLAN 3 and a router (R) with three virtual interfaces (one per VLAN) connected to an Ethernet link in trunk mode. PC1 and PC2 will establish two TCP connections to S with a very long duration (PC1 will be connected first and then PC2) and can generate data at a very high rate. Each of the virtual interfaces of router R has a buffer of 100 kB.





At t=0, PC1 establishes a TCP connection to S (PC2 is idle). The table shows the packet sequence monitored on PC1. The time is measured in milliseconds from t=0:

t (msec)	IP orig	IP dst	Port orig	Port dst	Flags	awnd
0	PC1	S	2000	80	SYN	100 kB
0.064	S	PC1	80	2000	SYN+ACK	16 kB
0.065	PC1	S	2000	80	ACK	100 kB

a) What would be the minimum RTT (round-trip time) we could have on this network for 3WHS packets if propagation delays are negligible and we do not take Ethernet headers into account? What is the buffer occupancy of R's interface in VLAN3 at t=0? (assume at that time on the network there could be traffic from other connections).

Minimum RTT:

Motivation:

Buffer occupancy:

Motivation:

b) Fill in the following table with the states of the TCP connection at PC1 and S for the indicated time instants. The last row of the table assumes that the third packet of the 3WHS has already been processed by its recipient.

t (msec)	TCP connection state at PC1	TCP connection state at S
0.010		
0.050		
0.110		

c) Once the connection is established, assuming now that there are no other connections on the network, PC1 starts a continuous data transfer to S. The packets are 1500 B. What do you think will be the maximum transfer rate it can reach? Who limits this rate, TCP flow

control/congestion or the capacity of the network links? What will be the average buffer occupancy of R's interface in VLAN3?

Maximum transfer rate:

Motivation:

Who limits the maximum transfer rate:

Buffer occupancy:

Motivation:

Once the PC1-S connection has reached its maximum speed, PC2 establishes a TCP connection to S, with the same awnd and packet size values. Once the connection is established, PC2 starts a continuous data transfer to S.

What do you think will be the maximum transfer rate between PC2 and S? Who limits that rate, TCP or the capacity of the network links? What effect will this new connection have on the transfer rate between PC1-S?

Maximum transfer rate:

Motivation:

What limits the máximum transfer rate:

Effect on PC1-S transfer:

We use TCP Tahoe, i.e. the version of TCP explained in class, with Slow-Start (SS) and Congestion Avoidance (CA). Assume the conditions of c).

e) If there have been no losses so far, what state will the congestion control mechanism of PC1 be in? What is the value of the variable ssthresh?.

State: ***ssthresh:***

A packet is now lost on the connection between PC1 and S (e.g., due to a transmission error).

f) What mechanism does TCP use to detect the loss? What will be the value of the congestion window and ssthersh right after the loss is detected? If there is no further loss afterwards, how much should the congestion window be worth to switch from SS to CA?

Detection of loss:

Congestion window and ssthresh:

Congestion window to switch from SS to CA:

Final exam Xarxes de Computadors (XC), Degree in Informatics Engineering		19/6/2024	Spring 2024
GIVEN NAME (UPPERCASE):	SURNAME (UPPERCASE):	GROUP:	DNI/NIE:

Duration: 3h total. The test will be collected in 30 minutes. Please answer on the same sheet.

Problem 3 (1,75 points)

Let's assume a new UPC webmail at URL: http://c.upc.edu/i.html and a browser with HTTP1.1 (with pipelining). Your laptop connects to the Internet at home. Your router does NAT and forwards DNS requests to your internet provider. RTT to this DNS server is 1 ms, and to any external DNS or HTTP server is 5 ms. Each content element (HTML, image) is 50 KB, MSS 1 KB, data transfer 1 MSS ~0 ms, awnd: 200 KB.

Your laptop and router at home are turned on (cache memory empty), and the browser is opened with that URL, which, therefore, must first resolve the DNS number in that URL.

a) What is the minimum time to receive the response of a DNS resolution? Indicate which servers are involved.

b) What is the maximum time to receive a response to a resolution? Indicate servers and DNS records (RR) involved.

For an immediate second query of the same DNS number on the web page visited: (e.g. the URL of an image)

c) When would the resolution be delayed, and why, assuming that the TTL of "c" was sufficiently long?

d) How the TTL in DNS data (RR) would affect, for example if TTL were 0 or 100?

What would be the page load time for the first visit to http://c.upc.edu/i.html (with DNS, TCP slow start, HTTP):

e) If the page does not include any images:

f) If the page includes 3 images on the same server: http://c.upc.edu/i1.png...i3.png and no limits on connection number:

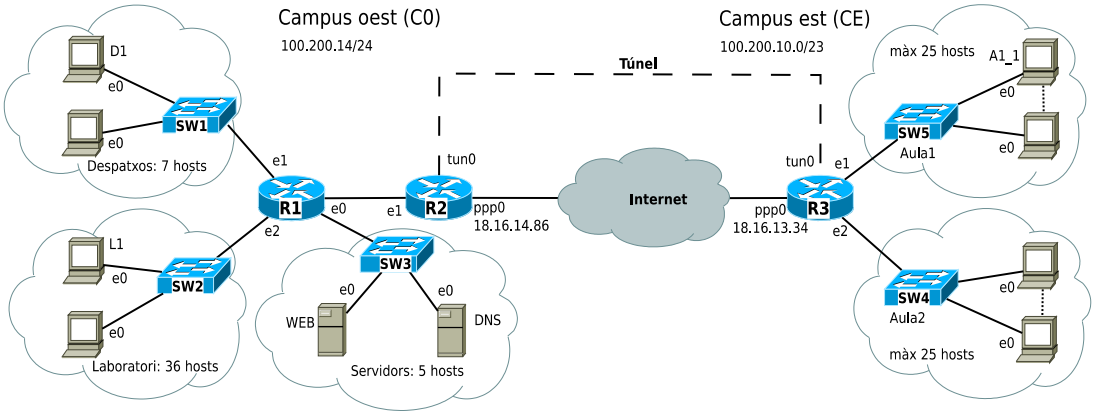
g) If the page includes 3 images on the same server and the browser only opens one connection per server:

h) If the 3 images are found at http://img.upc.edu/ and the browser can open unlimited connections:

Examen Final de Xarxes de Computadors (XC), Grau en Enginyeria Informàtica		19/06/2024	Spring 2024
FIRST NAME (CAPITALS):	**LAST NAME (CAPITALS):**	**GRUP:**	**DNI/ID:**

Duration: 3 hours. The quiz will be collected in 30 minutes. Answer the problems in the same sheet.

Problem 4 (2.5 points)
A university department has resources distributed between two campuses, the East (CE) and the West (CO). The attached figure shows the network infrastructure. The classrooms are in the CE. The offices, the laboratory, and the servers are in the CO.



There is a subnet for each switch, five in total: Classroom1, Classroom2, Offices, Laboratory, and Servers. The figure specifies the number of hosts for each one (example: Servers subnet has 5). Each host is connected to a switch. Each host in CE is assigned an IP from the range 100.200.10.0/23 and each one in CO from the range 100.200.14.0/24. All connections are Ethernet unless otherwise specified. All resources use the department's DNS server for name resolution.

Both sites are interconnected through the internet via an IP tunnel. The university's Internet Service Provider (ISP) has assigned the following network parameters to each of the campuses:

- **CO** Public IP: 18.16.14.86, gateway: 18.16.14.1
- **CE** Public IP: 18.16.13.34, gateway: 18.16.13.1

The uppercase letters of the resources indicate the IPs of the network cards; the lowercase letters indicate the MACs. For example, R1.e2 is the IP of the e2 interface of R1, while r1.e2 is its MAC. Use this notation unless the criteria are explicitly stated otherwise.

Answer the following questions using the free cells in the tables. Each answer must be consistent with the answers given in previous questions (addresses and IP ranges, etc.).

a) (0.25 points) Assign numeric IPs and masks in slash notation (e.g., /24) to R2.tun0 and R3.tun0, minimizing the number of unassigned addresses within the range and minimizing the use of public IPs.

IP R2.tun0	Mask	IP R3.tun0	Mask

b) (0.25 points) Complete the tunnel configuration on R3 so that CE accesses CO through the tunnel:

`R3# ip tunnel add` [ ] `mode gre remote` [ ] `local` [ ] `ttl 255`

`R3# ifconfig` [ ] [ ] `netmask` [ ]

`R3# route add` [ ] [ ] `netmask` [ ] `gw` [ ]

c) (0.25 points) Complete R3's route table to enable all CE hosts to access the internet through R2.

Destination name	Numeric prefix	Mask	Gateway	Interface
ISP-R3				

d) (0.25 points) How many classrooms can there be in CE knowing that each one can have up to 30 hosts?

Maximum number of classrooms	

e) (0.25 points) Assign a subrange of IPs to each subnet in CO such that the number of unassigned addresses within each subrange and between subranges is minimized. Order the rows of the table in ascending order by prefix. Indicate the subnet names, the number of assigned IPs, the prefix, and the subnet mask in slash notation.

Subnet name	Number of assigned IPs	Numeric prefix	Mask

f) (0.25 points) State whether NAT needs to be performed on any network element so that all hosts from both campuses can access each other and all can access the internet. If so, specify which ones.

Yes / No		**Which?**		**Why?**	

g) (0.25 points) Immediately after being restarted, the host with IP A1_1 from Classroom1 *pings exmple.com*, which is outside the university. Complete the sequence of IP datagrams that pass through R2.

Source IP	Destination IP	Protocol	Contents
A1_1			

h) (0.25 points) A1_1 performs a *traceroute* to the web server. Specify the devices and IP addresses that will appear on the list.

i) 0.25 points) Immediately after, the host with IP D1 from the offices is restarted and *pings* the DNS server. Complete the sequence of IP datagrams that pass through R1.

Ethernet			IP		
Source address	**Destination addr.**	**Contents**	**Source addre.**	**Destination ad.**	**Contents**
d1.e0					

j) (0.25 points) Specify the outbound (OUT) filtering rules that must be applied to R2.e1 so that from the Internet, access is exclusively allowed to WEB services (HTTP port 80 and HTTPS port 443) and DNS (port 53). These rules should not affect the hosts in CE.

Source IP	Source port	Destination IP	Destination port	Protocol	Action